Diaphragm Pump Design & Simulation Workflow

Engineering Design Steps, Inputs & Simulation Pipeline

1. REQUIREMENTS & PERFORMANCE (Initial Sizing)

Key Inputs:

- Required Flow Rate Q
- Maximum Operating Pressure P_{max}
- Vacuum / Suction Conditions $P_{vac/suc}$

Analytical Calculations:

- **Displacement Volume:** $V_d = \frac{Q}{N \cdot \eta_v}$
- **Stroke & Eccentric Radius:** $S = 2r \quad , \quad V_d = A_{eff} \cdot S$
- **Motor Power:** $P_{motor} = \frac{Q \cdot \Delta P}{\eta_{total}}$
- **Shaft Diameter:** $d \geq \left(\frac{16T}{\pi \tau_{allow}}\right)^{1/3}$
- **Valve Port Area:** $A_v = \frac{Q}{v_{max}}$

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2. MATERIAL SELECTION & STRUCTURAL FEA (ANSYS)

Inputs:

- Material Properties (EPDM, NBR, PTFE - Hyperelastic models)
- Boundary Conditions (Fixed outer rim, center deflection/force)

- **Hyperelastic Strain Energy Potential:** $W = W(I_1, I_2, I_3)$
- **Stress-Strain Relations:** $S_{ij} = \frac{\partial W}{\partial E_{ij}}$
- **Fatigue Life Prediction:** $N_f = C \cdot (\Delta\sigma)^{-\beta}$

Outputs / Evaluation:

- Stress & Strain Distribution
- Deflection Profile & Fatigue Life Analysis

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3. FLUID DYNAMICS (CFD)

Inputs:

- Fluid Properties (Density ρ, Viscosity μ)
- Dynamic / Moving Mesh – UDF in ANSYS FLUENT (Driven by diaphragm motion)

Formulas & Governing Equations:

- **Navier-Stokes Equations:** $\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}\right) = -\nabla P + \mu \nabla^2 \vec{v} + \vec{f}$
- **Continuity Equation:** $\nabla \cdot \vec{v} = 0$
- **Valve Pressure Drop:** $\Delta P_{valve} = \xi \cdot \frac{\rho v^2}{2}$

Outputs / Evaluation:

- Valve Pressure Drop & Flow Velocity Profiles
- Volumetric Efficiency Verification

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4. SYSTEM-LEVEL SIMULATION (MATLAB / Simulink - Simscape)

Block Components:

- **Electric Motor Equations:** $T_m = K_t \cdot I \quad , \quad V = IR + L\frac{dI}{dt} + K_e \omega$
- **Kinematic Linkage / Crank-Slider:** $x(t) = r\cos(\omega t) + \sqrt{l^2 - r^2 \sin^2(\omega t)}$
- **Check Valve Dynamics:** $m\ddot{x}_v + c\dot{x}_v + kx_v = A_v \Delta P(t)$
- **Instantaneous Mass Flow Rate:** $\dot{m}(t) = \rho \cdot A(t) \cdot \sqrt{\frac{2\Delta P(t)}{\rho}}$

Final Outputs:

- Characterization Curves
- Overall System Performance Optimization

Diaphragm Pump Design & Simulation Platform

An interactive engineering platform for the conceptual design, preliminary sizing, simulation, and optimization of diaphragm pumps for gas and liquid handling across diverse laboratory applications.

Tools: MATLAB (GUI Development) | Simulink / Simscape (System-Level Simulation) | ANSYS (Component-Level Simulation)

Feature	Gas Pump Platform	Liquid Pump Platform
Medium	Gas	Liquid
Functions	Aspiration, Compression, Transfer, Vacuum Generation	Aspiration, Dosing, Metering, Transfer
Design Inputs	Flow Rate, Outlet Pressure, Vacuum Level	Flow Rate, Pressure, Suction Height
Diaphragm	Diameter, Stroke, Material	Diameter, Stroke, Material
Valve	Valve Type, Valve Material	Valve Type, Valve Material
Sizing	Automatic / Manual	Automatic / Manual
Performance Analysis	Flow Rate, Pressure, Vacuum	Flow Rate, Pressure, Dosing Performance
Simulation	Motor-Driven & Diaphragm Motion	Motor-Driven & Diaphragm Motion
Optimization	Geometry & Operating Parameters	Geometry & Operating Parameters
Output	Preliminary Design & Performance Results	Preliminary Design & Performance Results

